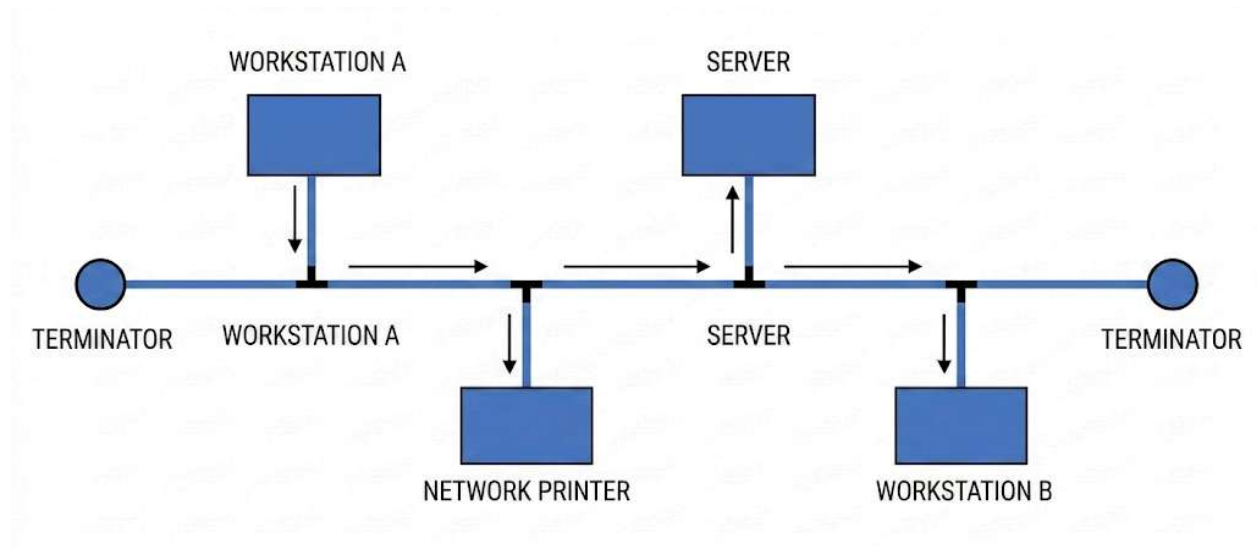
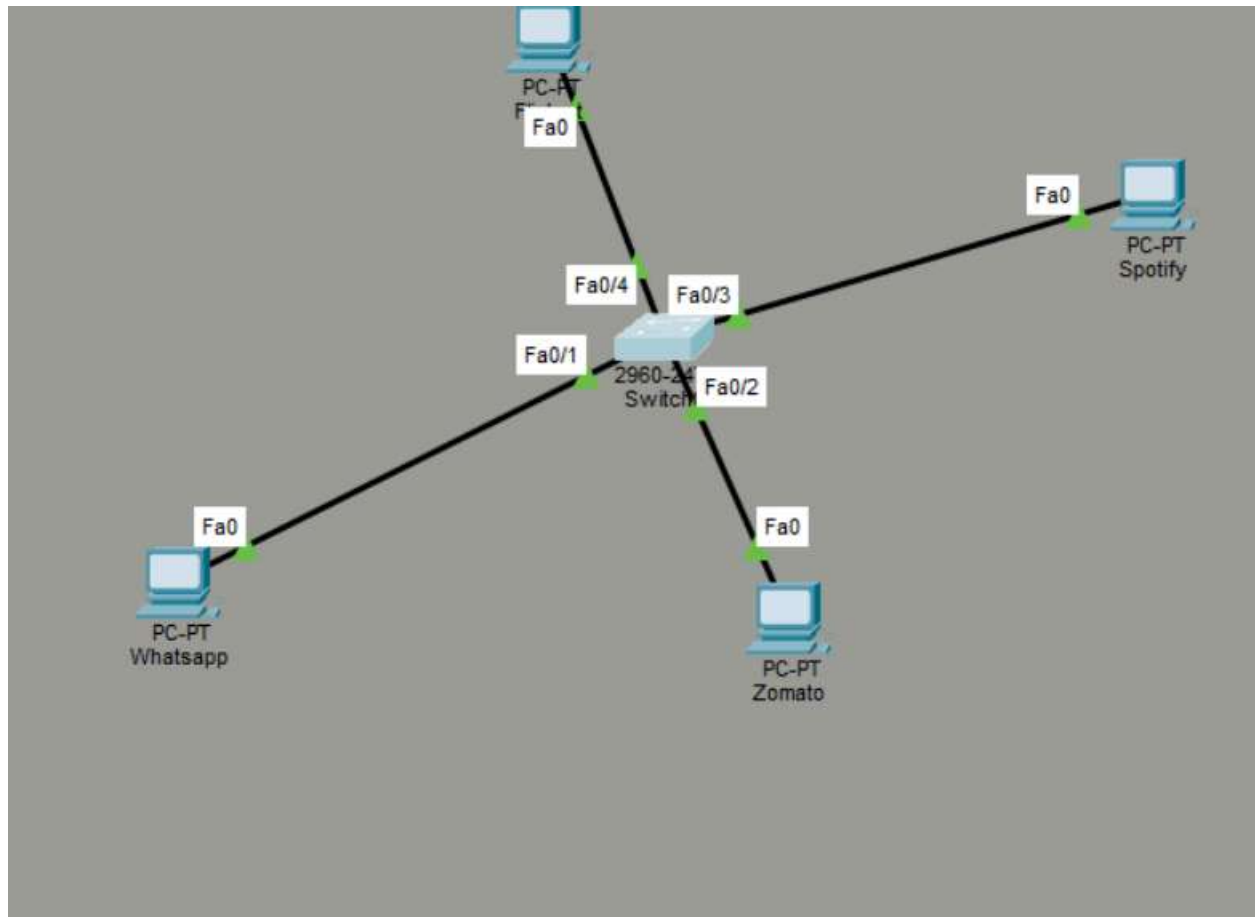


Task 1 : Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.



A real world example - used where there are only a few computers and peripherals need to communicate like in a small classroom, office. A primary real-world example of a bus topology in a professional or industrial setting is the **Controller Area Network (CAN bus)**.

Task 2 : In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.26

Pinging 192.168.10.26 with 32 bytes of data:

Reply from 192.168.10.26: bytes=32 time<1ms TTL=128
Reply from 192.168.10.26: bytes=32 time<1ms TTL=128
Reply from 192.168.10.26: bytes=32 time<1ms TTL=128
Reply from 192.168.10.26: bytes=32 time<1ms TTL=128

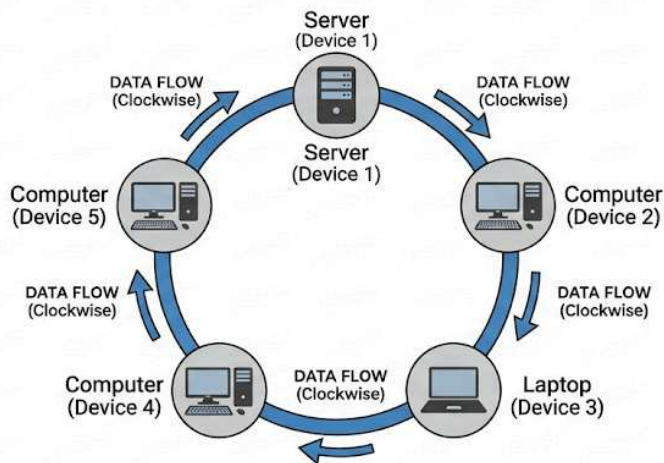
Ping statistics for 192.168.10.26:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Ping between PCs tried out.

Task 3: Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup. Hint: Use arrows to show the direction of data flow.

NETWORK RING TOPOLOGY DIAGRAM



In Ring Topology, each device is connected to two neighbors, forming a single circular network pathway. Data packets travel from one device to the next in one specific direction shown by arrows. Each device acts as a repeater receiving the data, amplifying it, and transmitting it to the next device in the ring until the packet reaches its intended destination.

Task 4: Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Topology	Advantages	Disadvantages
Mesh	1. High Reliability & Redundancy: If one link or node fails, traffic is automatically rerouted through alternate paths without disrupting the network. 2. Enhanced Security & Privacy: Dedicated channels between devices ensure data travels directly to its destination without public broadcasting.	1. High Cost: Requires a massive amount of cabling and hardware ports, making it expensive to deploy. 2. Complex Installation: Setting up and managing the extensive physical connections is difficult and labor-intensive.
Star	1. Easy to Expand & Manage: New devices can be added or removed simply by connecting or disconnecting them from the central hub. 2. Simple Fault Isolation: If a single cable or device fails, only that specific connection goes down; the rest of the network operates normally.	1. Single Point of Failure: If the central switch or hub crashes, the entire network goes offline. 2. Cable Dependent: High reliance on the central device means performance is bottlenecked by the central hub's capacity and requires longer individual cable runs to each node.

